

Technical Specifications · SYN-2026-001

TECHNICAL SPECIFICATIONS

- 6.1 Concrete. All structural concrete shall be of grade M30, manufactured and placed in accordance with IS 456:2000 Section 9. Cement shall conform to IS 269:2015 (Ordinary Portland Cement, 53 grade). Aggregate shall conform to IS 383:2016. Slump for placed concrete shall be 75 ± 25 mm measured at the point of placement.
- 6.2 Reinforcement steel. Reinforcement bars shall be high-strength deformed bars conforming to IS 1786:2008, grade Fe 500D, of nominal diameters as shown on the drawings.
- 6.3 Compaction. Subgrade compaction shall achieve a minimum dry density of 98% of MDD as per IS 2720 Part 7, measured by sand replacement method at intervals not exceeding 200 m of completed road length.
- 6.4 Bitumen. Bituminous concrete wearing course shall be 50 mm thick, laid at a temperature not less than 145 °C, complying with MoRTH Section 509.
- 6.5 Drainage. Side drains shall be of trapezoidal cross-section with bed width 0.5 m, side slopes 1:1.5, and minimum longitudinal gradient of 0.3%.
- 6.6 Curing. Concrete shall be cured by ponding or wet hessian for a minimum of 7 days from placement, kept continuously moist throughout the curing period.
- 6.7 Aesthetic finish. The finished surface shall present an aesthetically pleasing appearance and be free from visible defects to the extent practicable.